

E-Commerce Sales Analysis

1. Project Overview

This project focuses on analyzing e-commerce sales data to understand sales performance, product trends and revenue patterns. Python was used for data cleaning and exploratory analysis, SQL was used for data analysis and power BI was used to create an interactive dashboard.

2. Problem Statement

Identify which item types blinkit should prioritize to maximize sales revenue by analyzing the sales performance of different product categories and determine which outlet format blinkit should prioritize for expansion by analyzing the revenue generated by each outlet type.

3. Dataset description

- **Dataset Overview**

This project uses a Blinkit Grocery Sales dataset containing transactional and product-level information from Blinkit's e-commerce grocery business. The dataset is used to analyze sales performance, customer ratings, outlet characteristics, and product categories to generate business insights.

- **Dataset Summary**

Dataset Name : Blinkit Grocery Sales Dataset

Domain : E-commerce / Retail

Records : 8,523

Columns : 12

Data Type : Structured CSV

4. Exploratory Data analysis

- **Data Cleaning using Python**

- ◆ Checked missing values
- ◆ Removed duplicate records
- ◆ Corrected data types of attributes
- ◆ Standardized categorical Values
- ◆ Checked invalid values
- ◆ Feature engineering

- **Explore the data using Python**

- ◆ Category Performance
- ◆ Outlet type Performance
- ◆ Outlet Location type Performance
- ◆ Outlet Size performance
- ◆ Sales per Outlet type performance

5. Data Analysis using SQL

We performed structured analysis in MySQL to answer business problem.

1. **Outlet location type** : we find which outlet location type perform the best.

Result Grid



Filter Rows:

Export:



Wrap Cell Content:

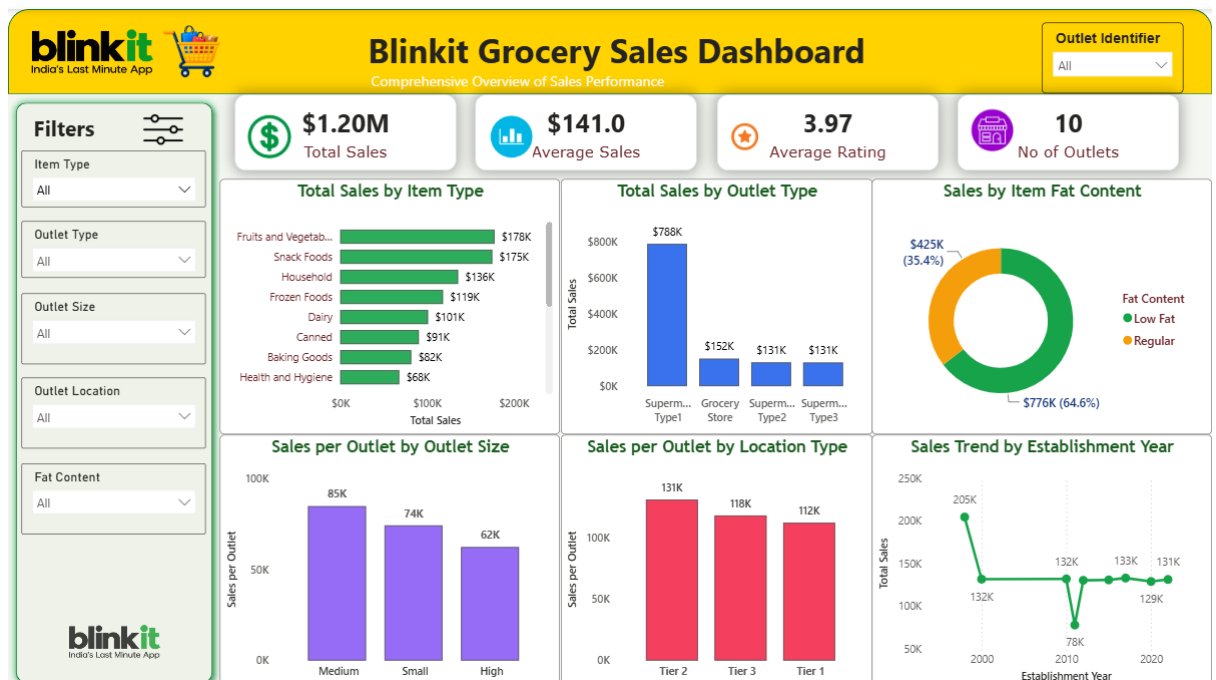


	outlet_location_type	number_of_outlets	sales	sales_per_outlet
▶	Tier 2	3	393150.64	131050.21
	Tier 3	4	472133.03	118033.26
	Tier 1	3	336397.81	112132.6

- Total Revenue** : Total revenue generated by all orders.

Result Grid		Filter Rows:
	Total_Revenue	
▶	1201681	

6. Power BI Dashboard



7. Business Recommendations

- Expand Supermarket Type1 Outlets**
Prioritize opening more Supermarket Type1 stores, as they generate the highest overall revenue.
- Prioritize Fruit and Vegetables inventory**
Maintain a high and consistent inventory of Fruits & Vegetables because this category generates the highest total sales and experiences strong customer demand. Ensure frequent restocking and prioritize shelf availability to prevent stock-outs.
- Increase Investment in Tier 2 Outlets**

Prioritize expansion and inventory allocation for Tier 2 outlets. Increase stock availability of high-demand products, launch targeted local promotions, and consider opening additional outlets in similar Tier 2 markets.

4. Focus on High-Performing Outlet Formats

Blinkit should prioritize expanding the outlet formats that deliver the highest sales per outlet: **Small-sized Supermarket Type 1 in Tier 1, High-sized Supermarket Type 1 in Tier 3, and Medium-sized Supermarket Type 2 in Tier 3.** These formats demonstrate the strongest outlet productivity and revenue potential.